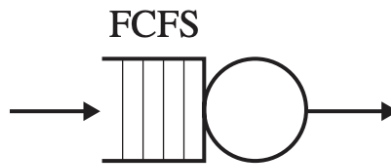


2.2 Slowdown

(a) Jobs arrive at a server that services them in FCFS order:



The average arrival rate is $\lambda = \frac{1}{2}$ job/sec. The job sizes (service times) are independently and identically distributed according to random variable S where

$$S = \begin{cases} 1 & \text{with probability } 3/4 \\ 2 & \text{otherwise} \end{cases}.$$

You have measured the mean response time: $\mathbf{E}[T] = \frac{29}{12}$.

Based on this information, compute the mean slowdown, $\mathbf{E}[\text{Slowdown}]$, where the slowdown of job j is defined as $\text{Slowdown}(j) = \frac{T(j)}{S(j)}$, where $T(j)$ is the response time of job j and $S(j)$ is the size of job j .

(b) If the service order in part (a) had been Shortest-Job-First (SJF), would the same technique have worked for computing mean slowdown?

(a) $E[S] = 1 \cdot 3/4 + 2 \cdot 1/4 = 5/4$
 $E[\text{Slowdown}] = E[T] / E[S] = 29/15$

(b) No, the same technique won't work for SJF.

In FCFS, we can calculate the delay rate by directly relating the delay time $T(j)$ of each task to its size $S(j)$.

However, in the case of Shortest-Job-First (SJF), the service order changes according to the size of the tasks, and the smaller tasks are prioritized. This results in the relationship between the response time $T(j)$ and the size $S(j)$ of each task no longer being fixed.

Therefore, it is not possible to directly apply the FCFS method to calculate the average latency.

6.2 Simplified Power Usage in Server Farms

Given that power is expensive, it is common practice to leave servers on only when they are being used and to turn them off whenever they are not in use. Assume that the following power-aware algorithm is used: When a job arrives, it instantly turns on a fresh server (assume zero setup cost). When the job completes service, it instantly turns off that server. Assume that there is always a server available for every job (i.e., there is no queueing). Your goal is to derive the time-average rate at which power is used in our system. Assume that when a server is on, it consumes power at a rate of $\mathcal{P} = 240$ watts. Assume $\lambda = 10$ jobs arrive per second and that the service requirement of jobs is Uniformly distributed ranging from 1 second to 9 seconds.

(Note: This is a highly simplified model. We will study much more complex models in Chapter 27.)

Average service requirement time = $1/9 \sum_{i=1}^{10} i = 55/9$ seconds

Time-average rate = $240 * 55/9 \approx 1466.67$ watts

8.1 Solving for Limiting Distribution

Consider the program analysis problem from Section 8.3.3. Determine the limiting distribution, (π_C, π_M, π_U) , by solving the stationary equations.

In 8.3.3.

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.8 & 0.1 & 0.1 \\ 0.9 & 0.1 & 0 \end{bmatrix}$$

$$\text{We know, (1) } [\pi_c \ \pi_M \ \pi_v] \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.8 & 0.1 & 0.1 \\ 0.9 & 0.1 & 0 \end{bmatrix} = [\pi_c \ \pi_M \ \pi_v]$$

$$(2) \pi_c + \pi_M + \pi_v = 1$$

$$\text{So, } \begin{cases} 0.7\pi_c + 0.8\pi_M + 0.9\pi_v = \pi_c & \text{--- (1)} \\ 0.2\pi_c + 0.1\pi_M + 0.1\pi_v = \pi_M & \text{--- (2)} \\ 0.1\pi_c + 0.1\pi_M = \pi_v & \text{--- (3)} \\ \pi_c + \pi_M + \pi_v = 1 & \text{--- (4)} \end{cases}$$

$$\begin{aligned} \text{Substitute (3) into (1)} &\Rightarrow 0.89\pi_M = 0.21\pi_c \\ &\Rightarrow \pi_M = 0.236\pi_c \quad \text{--- (5)} \end{aligned}$$

$$\text{Substitute (3) into (2)} \Rightarrow \text{same as (5)}$$

$$\text{Substitute (3) into (4)} \Rightarrow 1.1\pi_c + 1.1\pi_M = 1 \quad \text{--- (6)}$$

$$\begin{aligned} \text{Substitute (5) into (6)} &\Rightarrow 1.1\pi_c + 0.26\pi_c = 1 \\ &\Rightarrow \pi_c = 0.735 \quad \text{--- (7)} \end{aligned}$$

$$(7) \rightarrow (5) \Rightarrow \pi_M = 0.236 \cdot 0.735 = 0.174 \quad \text{--- (8)}$$

$$(7), (8) \rightarrow (3) \Rightarrow \pi_v = 0.0735 + 0.0174 = 0.091$$

$$\text{Ans : } (\pi_c, \pi_M, \pi_v) = (0.735, 0.174, 0.091)_{\#}$$

8.6 Threshold Queue

We define a threshold queue with parameter T as follows: When the number of jobs is $< T$, then the number of jobs decreases by 1 with probability 0.4 and increases by 1 with probability 0.6 at each time step. However, when the number of jobs increases to $> T$, then the reverse is true, and the number of jobs increases by 1 with probability 0.4 and decreases by 1 with probability 0.6 at each time step, as shown in Figure 8.8.

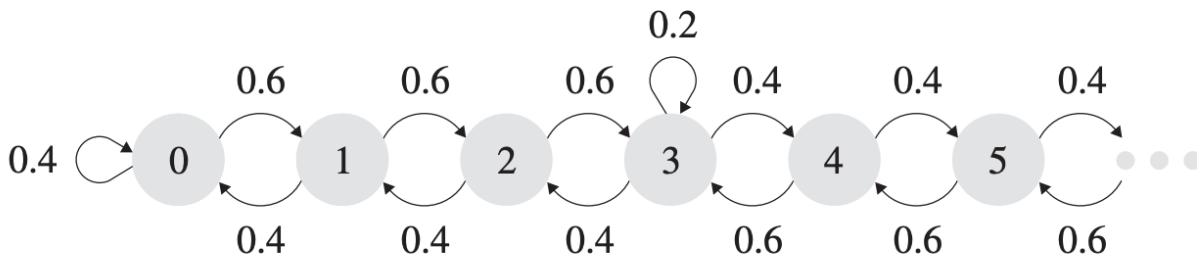


Figure 8.8. Markov chain for threshold queue with $T = 3$.

- (a) Assume that the limiting probabilities exist. Use the stationary equations to derive the limiting probability distribution as a function of T , for arbitrary threshold T .
- (b) Compute the mean number of jobs, $\mathbf{E}[N]$, in a threshold queue as a function of T .
- (c) What happens to $\mathbf{E}[N]$ when $T = 0$? Does this answer make sense?

(a)

Limiting probability : $\sum_{i=0}^{\infty} \pi_i = 1$

Assume $\pi = [\pi_0 \ \pi_1 \ \dots]$, $\pi P = \pi$

$$\text{And matrix } P = \begin{bmatrix} 0.4 & 0.6 & 0 & \dots & 0 & 0 & \dots \\ 0.4 & 0 & 0.6 & \dots & 0 & 0 & \dots \\ 0 & 0.4 & 0 & & 0.6 & 0 & \dots \\ \vdots & \vdots & \vdots & & 0.2 & 0.4 & \dots \\ \vdots & \vdots & \vdots & & 0.6 & 0 & \dots \\ 0 & 0 & 0 & \dots & 0 & 0.6 & \dots \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \ddots \end{bmatrix}$$

$$\pi P = \pi$$

$$\Rightarrow [\pi_0 \ \pi_1 \ \dots] P = [\pi_0 \ \pi_1 \ \dots]$$

$$\Rightarrow 0.6 \pi_{i-1} + 0.4 \pi_{i+1} = \pi_i, \text{ if } i < T.$$

$$0.6 \pi_{i-1} + 0.2 \pi_i + 0.6 \pi_{i+1} = \pi_i, \text{ if } i = T.$$

$$0.4 \pi_{i-1} + 0.6 \pi_{i+1} = \pi_i, \text{ if } i > T.$$

$$\text{For } i \leq T, \quad 0.6 \pi_{i-1} = 0.4 \pi_i \Rightarrow \pi_i = \frac{3}{2} \pi_{i-1}$$

$$\Rightarrow \pi_i = \left(\frac{3}{2}\right)^i \pi_0$$

$$\text{For } i > T, \quad 0.4 \pi_{i-1} = 0.6 \pi_i \Rightarrow \pi_i = \frac{2}{3} \pi_{i-1}$$

$$\Rightarrow \pi_i = \left(\frac{2}{3}\right)^i \pi_T$$

$$\text{Finally, } \sum_{i=0}^{\infty} \pi_i = 1 \Rightarrow \pi_0 + \pi_1 + \dots + \pi_T + \pi_{T+1} + \dots = 1$$

$$\Rightarrow \pi_0 \left[1 + \frac{3}{2} + \left(\frac{3}{2}\right)^2 + \dots + \left(\frac{3}{2}\right)^{T-1}\right] + \pi_T \left[1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots\right] = 1$$

Let's simplify the equation:

$$\text{First : } 1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots = \frac{1}{1 - \frac{2}{3}} = 3$$

$$\text{Second : } 1 + \frac{3}{2} + \left(\frac{3}{2}\right)^2 + \dots + \left(\frac{3}{2}\right)^{T-1} = \frac{1 - \left(\frac{3}{2}\right)^T}{1 - \frac{3}{2}} = 2 \left[\left(\frac{3}{2}\right)^T - 1 \right]$$

$$\text{Third : } \pi_T = \left(\frac{3}{2}\right)^T \pi_0$$

$$\text{So the equation becomes } 2 \left[\left(\frac{3}{2}\right)^T - 1 \right] \pi_0 + 3 \cdot \left(\frac{3}{2}\right)^T \pi_0 = 1$$

$$\text{And } \pi_0 = \left[5 \left(\frac{3}{2}\right)^T - 2 \right]^{-1}$$

$$\begin{aligned} (b) \quad E[N] &= \sum_{n=0}^{\infty} n \cdot \pi_n = \sum_{n=0}^T n \cdot \left(\frac{2}{3}\right)^n \pi_0 + \sum_{n=T+1}^{\infty} n \cdot \left(\frac{2}{3}\right)^{n-T} \pi_T \\ &= \pi_0 \cdot \sum_{n=0}^T n \cdot \left(\frac{2}{3}\right)^n + \left(\frac{3}{2}\right)^T \pi_0 \cdot \sum_{n=T+1}^{\infty} n \cdot \left(\frac{2}{3}\right)^{n-T} \end{aligned}$$

(c) When $T = 0$, no matter how many jobs in queue, the probability of decreases by 1 will always be 0.6, and the probability of increases by 1 will always be 0.4.

In this condition, $E[N]$ will be small, representing that the queue will empty quickly or remain at a low number of tasks, so it is make sense.

10.3 Time to Empty

Consider a router where, at each time step, the number of packets increases by 1 with probability 0.4 and decreases by 1 with probability 0.6. We are interested in the time required for the router to empty. The Markov chain depicting the number of packets is shown in Figure 10.8. Let $T_{1,0}$ denote the time to get

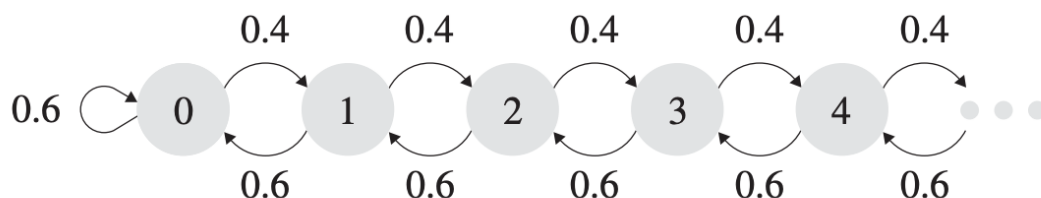


Figure 10.8. Number of packets at router.

from state 1 to state 0. (a) Compute $\mathbf{E}[T_{1,0}]$. (b) Compute $\mathbf{Var}(T_{1,0})$. [Hint: The variance computation is a little tricky. Be careful not to lump together distinct random variables.]

Since the number of packets decreases by 1 is with probability 0.6, we can get

$$E[T_{1,0}] = \frac{1}{0.6} = 5/3.$$

11.3 Doubling Exponentials

Suppose that job sizes are Exponentially distributed with rate μ . If job sizes all double, what can we say about the distribution of job sizes now? Prove it.

If the job sizes are exponentially distributed with μ , then PDF of job size X is $f_X(x) = \mu e^{-\mu x}$, $x \geq 0$.

We use Integration by substitution :

$$\text{If } y = g(x), \text{ then } f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Assume the new job size is Y , and $Y = 2X$.

$$\begin{aligned} \Rightarrow f_Y(y) &= f_X\left(\frac{y}{2}\right) \cdot \left| \frac{d}{dy} \left(\frac{y}{2}\right) \right| = \mu e^{-\mu \cdot (y/2)} \cdot \left| \frac{1}{2} \right| \\ &= \frac{1}{2} \mu e^{-\frac{\mu}{2} \cdot y} \end{aligned}$$

11.5 Practice with Definition of Poisson Process

- (a) Consider a stream of packets arriving according to a Poisson process with rate $\lambda = 50$ packets/sec. Suppose each packet is of type “green” with probability 5% and of type “yellow” with probability 95%. Given that 100 green packets arrived during the previous second, (i) what is the expected number of yellow packets that arrived during the previous second? (ii) what is the probability that 200 yellow packets arrived during the previous second?
- (b) Red packets arrive according to a Poisson process with rate $\lambda_1 = 30$ packets/sec. Black packets arrive according to a Poisson process with rate $\lambda_2 = 10$ packets/sec. Assume the streams are statistically multiplexed into one stream. Suppose we are told that 60 packets arrived during the second. What is the probability that exactly 40 of those were red?
- (c) Suppose packets arrive according to a Poisson process with rate λ , and you are told that by time 30 seconds 100 packets have arrived. What is the probability that 20 packets arrived during the first 10 seconds?

(a)(i) 100 green packets arrived during previous second, packet with "green" is 5%

So estimated packets arrived during previous second N is

$$N = 100 \div 0.05 = 2000$$

And the number of yellow packet is $2000 - 100 = 1900$ #

(ii) $\lambda = 50$, and 95% are yellow $\Rightarrow$ The number of yellow packet arrived every second is $\lambda_y = 50 \times 0.95 = 47.5$.

And we describe "probability of 200 yellow packets arrived during previous second" as $P(A(t+1) - A(t) = 200)$

$$P(A(t+1) - A(t) = 200) = e^{-47.5} \cdot \frac{(47.5)^{200}}{200!} \quad \#$$

(b) Assume total packet arrive with rate λ ,

$$\lambda = \lambda_1 + \lambda_2 = 40 \text{ packets/sec.}$$

With binomial distribution, $n=60$, $p = \frac{\lambda_1}{\lambda} = \frac{3}{4}$, n is the number of packets arrived during the second, and p is the probability of the arrived packet is red.

$$\text{So } P(40 \text{ packets are red}) = \binom{60}{40} \left(\frac{3}{4}\right)^{40} \left(\frac{1}{4}\right)^{20} \quad \#$$

(c) With binomial distribution, $n=100$, $p = \frac{10}{30} = \frac{1}{3}$, n is the number of packets arrived in 30 second, and p is the probability of the arrived packet is in first 10 second.

$$\text{So } P(20 \text{ packet arrived in first 10 sec}) = \binom{100}{20} \left(\frac{1}{3}\right)^{20} \left(\frac{2}{3}\right)^{80} \quad \#$$

11.7 Malware and Honeypots

A new malware is out in the Internet! Our goal is to estimate its spread/damage by time t , assuming it starts at time 0. Assume that the Internet hosts get infected by this malware according to a Poisson process with parameter λ , where λ is *not known*. Thrasyvoulos installs a Honeypot security system to detect whether hosts are infected. Unfortunately there is a *lag time* between when a computer is infected by the malware and the Honeypot detects the damage. Assume that this lag time is distributed $\text{Exp}(\mu)$. Suppose that Thrasyvoulos's Honeypot system has detected $N_1(t)$ infected hosts by time t . Thrasyvoulos worries that, because of the lag, the number of infected hosts is actually much higher than $N_1(t)$. The goal of the problem is to understand how many *additional* hosts, $N_2(t)$, are expected to also be infected at time t .

- (a) Suppose that an infection happens at time s , where $0 < s < t$. What is the probability that the infection is detected by time t ?
 - (b) Consider an arbitrary infection that happens before time t . What is the (unconditional) probability, p , that the infection is detected by the Honeypot by time t ?
 - (c) How can we use our knowledge of $N_1(t)$ to estimate λ ?
 - (d) Use your estimate of λ to determine the expected value of $N_2(t)$.
- [Note: None of the above solutions requires more than a couple lines.]

(a) We define lag time as LT , if an infection happens at time s , and be detected by time t , means $s + LT \leq t$.

$\Rightarrow$ We need to know $P\{LT \leq t - s\}$

The CDF of exponential distribution with μ is

$$P\{X \leq s\} = 1 - e^{-\mu s}.$$

$$\text{So } P\{LT \leq t - s\} = 1 - e^{-\mu(t-s)} \quad \#$$

(b) Assume the time of infection is uniform distribution in $[0, t]$.

$$\text{Then } p = \frac{1}{t} \int_0^t [1 - e^{-\mu(t-s)}] ds.$$

(c) The host get infected with λ , but the infected hosts get detected with probability p , so $N_1(t)$ follow the Poisson distribution with λp .

$$\text{Means that } N_1(t) = (\lambda p) \cdot t \Rightarrow \lambda = \frac{N_1(t)}{pt} \quad \#$$

(d) Assume the number of infected hosts is $N(t)$, since $N(t) = \lambda t$. The expected value of $N_2(t)$ is

$$E[N_2(t)] = E[N(t)] - E[N_1(t)] = \lambda t - \lambda p t = \lambda \cdot t(1-p)$$

$$= \frac{N_1(t)}{p} \cdot (1-p) = N_1(t) \left(\frac{1-p}{p} \right) \quad \#$$

13.2 Server Farm

In the server farm shown in Figure 13.7, jobs arrive according to a Poisson process with rate λ and are probabilistically split between two servers, with p fraction of the jobs going to server 1, which has service rate μ_1 , and $q = 1 - p$ fraction going to server 2, which has service rate μ_2 . Assume that job sizes are Exponentially distributed. Derive an expression for the mean response time, $E[T]$, experienced by arrivals to the server farm.

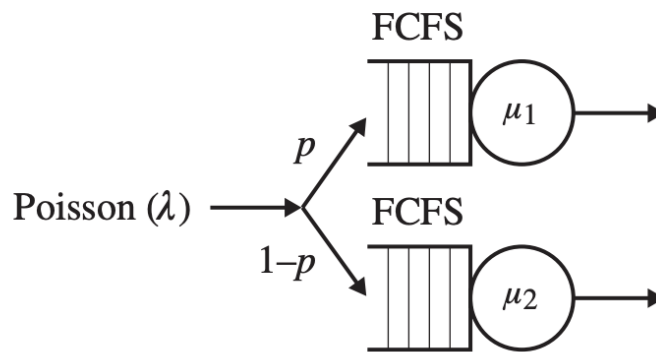


Figure 13.7. Server farm from Exercise 13.2.

The arrival rate of server 1 is $\lambda_1 = \lambda p$

The arrival rate of server 2 is $\lambda_2 = \lambda(1-p)$

We can consider this server farm as 2 M/M/1 queueing system

1st queueing system : arrival rate = λp , service rate = μ_1

2nd queueing system : arrival rate = $\lambda(1-p)$, service rate = μ_2

The response time in M/M/1 is $T = \frac{1}{\mu - \lambda}$

$$\Rightarrow E[T_1] = \frac{1}{\mu_1 - \lambda p}, \quad E[T_2] = \frac{1}{\mu_2 - \lambda(1-p)}$$

$$E[T] = \frac{p}{\mu_1 - \lambda p} + \frac{1-p}{\mu_2 - \lambda(1-p)}$$

13.10 Busy Period in M/M/1⁴

A busy period, B , is the time from when the system has 1 job until the system is first empty. (Obviously, the number of jobs may go up and down a lot in going from state 1 to state 0.) Your goal is to derive $\mathbf{E}[B]$ for an M/M/1.

- (a) Draw the CTMC for the M/M/1 with arrival rate λ and service rate μ .
- (b) Write a recurrence relation for $\mathbf{E}[B]$ by conditioning on the next state that the CTMC moves to after leaving state 1.
- (c) Solve the recurrence relation for $\mathbf{E}[B]$. What does the expression for $\mathbf{E}[B]$ remind you of?

(a)

